

SHLOK PANCHAL

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EXPERIENCE

Data Scientist · Vijay Exports Co. In · Mumbai, Maharashtra

July 2025 – Present

- Cut manual image lookup time by 70% by designing and deploying a production visual search engine over 600K+ product images using CLIP ViT-B/32 embeddings and FAISS IVF indexing as the live inventory search layer.
- Reduced retrieval latency by 65% and improved recall on visually similar, mismatched-label products by replacing keyword-based exact-match lookup with Approximate Nearest Neighbor (ANN) retrieval.
- Reduced irrelevant search results by an estimated 40% by engineering a SQLite metadata layer linking FAISS vector indices to SKU IDs and product attributes for structured pre-filtering alongside semantic retrieval.

Lead Business Analyst Intern · Ozibook Tech Solution Pvt. Ltd. · Bengaluru, Karnataka

January 2024 – March 2024

- Drove a 15% improvement in customer conversion rate by developing Tableau funnel dashboards for 3 product lines that identified conversion drop-off points.
- Contributed to an 8% uplift in campaign ROI by conducting statistical audience segmentation on engagement data across 5+ campaigns using Python (Pandas, NumPy).

PROJECTS

Enterprise Image Search — Android App & REST API

Flutter · Dart · ASP.NET Core 8 · SQLite · JWT

- Built a full-stack Android application (Flutter frontend, ASP.NET Core 8 REST API) enabling teams to instantly search a self-hosted 600K+ file NAS image library by SKU/product code, with thumbnail preview, batch export (ZIP + Excel), and folder browsing.
- Supported batch lookups of up to 50,000 SKU codes and instant search of up to 100 codes by designing a SQLite-backed indexing pipeline (Dapper) with background crawlers, incremental sync, wildcard matching, and zone-based filtering.
- Secured the platform for enterprise deployment by implementing JWT authentication with a device-approval workflow, refresh-token rotation, BCrypt password hashing, audit logging, and role-based access control (Admin/User).
- Improved mobile responsiveness by building an async, cancellable job queue (Channel-based workers) for ZIP/Excel generation and a multi-tier thumbnail cache (memory + disk), alongside biometric/PIN app lock and offline job history via local Isar persistence.

SmartPark AI — Autonomous Parking Agent

Python · Unity ML-Agents · PPO + GAIL · TensorFlow · Optuna

- Reduced early-phase exploration time by 40% versus a reward-only PPO baseline by training a PPO + GAIL reinforcement learning agent across 4 parking configurations in a Unity 3D simulation, using GAIL warm-start from expert demonstrations.
- Designed a shaped reward function and a camera-based perception module emulating real-world sensor input, architecting a CAN bus integration layer for forward-compatible real-vehicle deployment.

EDUCATION

Shah and Anchor Kutchhi Engineering College — B.Tech. in Artificial Intelligence & Data Science

October 2022 – May 2025

K.J. Somaiya Polytechnic — Diploma in Computer Engineering

October 2019 – August 2022

LANGUAGES & TECHNOLOGIES

Mobile & Full-Stack: Flutter, Dart, Android, ASP.NET Core 8, C#, REST API Design, JWT Auth, SQLite, Dapper, Provider (MVVM), Dio, Isar

Backend & Data: Python, SQL (MySQL, SQLite), Pandas, NumPy, AWS, GitHub, Jupyter

AI/ML: Deep Learning, Computer Vision, NLP, Reinforcement Learning, ANN/Vector Search, FAISS, PyTorch, TensorFlow